

Radio Frequency Fingerprinting for WiFi Devices Using Oscillator Drifts

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Abstract—Radio frequency fingerprint (RFF) identification is a promising technique that exploits hardware impairments-induced features to achieve specific device identification. Among RFF features, carrier frequency offset (CFO) as a hotspot feature has received widespread attention. Since CFO is time-variant, existing research suggests compensating for its drift; however, this paper emphasizes utilizing the drift of CFO. Correspondingly, a novel RFF feature, named cyclic similarity (cyc-similarity), is proposed to depict the oscillator drift. Simply combined the cyc-similarity feature with a K-nearest neighbor (KNN) classifier, the system can achieve superior temporal and receiver generalization performance. On a public dataset of WiFi devices, the proposed method outperforms the existing methods.

Index Terms—CFO drift, cyclic similarity, device identification, receiver robust.

I. INTRODUCTION

RADIO frequency fingerprint (RFF) originates from hardware defects introduced during the manufacturing process, which is unique and difficult to impersonate [1]. Hence, even if devices of the same type transmit exactly the same radio frequency (RF) signal, they can be distinguished by RFF features [2]. For every wireless device, the oscillator is a significant component, which is utilized for the conversion between baseband and RF signals. Thus, the extraction of oscillator-related impairment features has received extensive attention for the design of RFF identification (RFFI) systems.

Indeed, oscillator-induced impairment is the carrier frequency offset (CFO) in RF signals, which can be used as the RFF feature. Considering the differences in oscillators between different devices, authors in [3], [4] estimate the CFO using channel state information measurements, and use it as a long-term device fingerprint. On this basis, authors in [5] further extract five features for device identification, including coarse CFO, fine CFO, symbol timing synchronization, and amplitude and phase imbalance. Unfortunately, since the oscillator is sensitive to temperature fluctuations, the CFO is a time-variant impairment feature [6]. This results in unstable performance of the RFFI system that solely utilizes CFO for identification. Hence, CFO compensation is widely adopted to mitigate its variation before extracting RFF features [6]–[10]. However, different from the existing work, this paper finds that compensating the CFO may not bring improvements. On the contrary, although the oscillator varies over time, its drift can

become a unique RFF feature. The contribution of this paper is summarized as follows.

- We are the first to utilize oscillator drifts for device identification. Correspondingly, a novel RFF feature, named cyclic similarity (cyc-similarity), is proposed to represent this drift. Besides, this feature is extracted directly from raw preambles, avoiding redundant signal processing.
- The proposed RFF feature can well characterize the impairment feature of the device, so an excellent performance is achieved using a K-nearest neighbor (KNN) classifier. Besides, our RFFI system is not sensitive to the choice of the number of neighbors.
- An open dataset containing 6 WiFi devices operating on 4 days and 12 receivers is utilized to evaluate the proposed system, which achieves good performance in terms of temporal and receiver generalization.

The remaining sections of this paper are organized as follows. Section II briefly introduces the selected dataset. In Section III, the cyc-similarity is detailed, followed by the experimental results in Section IV. Finally, Section V concludes this paper.

II. PRELIMINARY

A. Signal of Interest

In a practical scenario with limited collection conditions, only signals of devices collected from a single receiver on a single day are available for the design of an RFFI system. While in practical applications, we expect that this system is able to identify signals collected from different days and different receivers. To perform verification in this scenario, signals need to be collected from different receivers on different days. Fortunately, such an experiment was deployed on the orbit testbed to collect WiFi signals on different days from different receivers, and the corresponding WiSig dataset was made available [10]. Specifically, the selected WiSig contained 6 WiFi devices using IEEE 802.11g standard with a center frequency of 2,462 MHz and a bandwidth of 20 MHz, and 12 different receivers of 25 Msps with the center frequency of 2,462 MHz, and the collections were performed on 4 days. For each device, more than 1,000 bursts were captured per day. In line with [6]–[10], we only utilize preambles to prevent the use of protocol-dependent and data-dependent knowledge.

B. CFO Drifts Across Different Days and Receivers

Fig. 1 shows the devices with smaller CFO differences across different days and receivers. Intuitively, CFO is time-varying, and it is difficult to identify them using the CFO alone. It is observed that for the same receiver, the CFO

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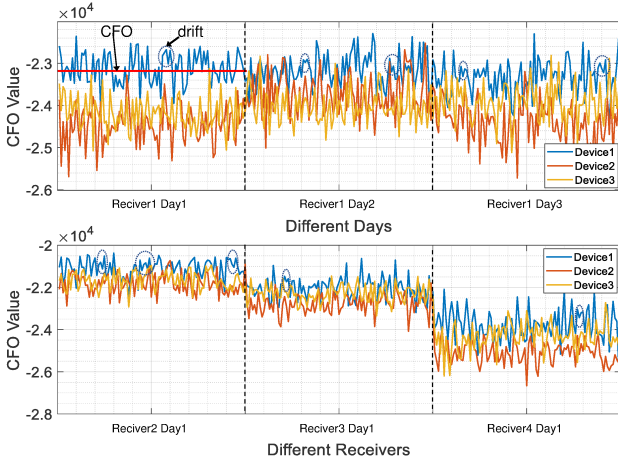


Fig. 1: CFO drifts across different days and receivers.

of devices varies less on different days, while for different receivers that varies relatively widely. The variation of CFO poses a significant challenge to the temporal and receiver generalization performance of the RFFI system [9].

Next, we detail the instantaneous phase of an oscillator, which can be expressed by [11]

$$\varphi(t) = 2\pi(f - f_n)t + w(t), \quad (1)$$

where f and f_n respectively represent the true frequency and the nominal frequency, and its difference is the CFO of the oscillator. $w(t)$ is a zero-mean stationary phase drift process, which can be modeled as a Wiener process [11], as follows

$$w(t) = w(t-1) + \delta(t), \quad (2)$$

where $\delta(t)$ is the phase increment. It is worth noting that the statistics of $w(t)$ depend on the hardware implementation. Hence, this paper aims to design the unique RFF feature to depict the characteristic of the oscillator drifts $w(t)$, which can be used for device identification.

III. CYC-SIMILARITY FEATURE ENGINEERING

Notably, each transmission burst of the IEEE 802.11g WiFi signal must begin with a distinctive preamble that is comprised of short training sequences (STS) and long training sequences (LTS). For ease of description, we mainly use the STS for detailed analysis. Let $x_S(m, n)$ ($n = 1, \dots, N_S$, with $N_S = 16$, and $m = 1, \dots, M_S$, with $M_S = 10$) denote the n -th sample of the m -th STS. The signal $x_S(m, n)$ is the ideal preamble for transmission, and the corresponding received signal is represented by $y_S(m, n)$.

At the transmitter, $x_S(m, n)$ is finally up-converted into the RF signal through an oscillator as follows,

$$y_{\text{rf}}(m, n) = x_S(m, n) e^{j(2\pi(f_t - f_n)t + w_t(t))}, \quad (3)$$

where f_t and $w_t(t)$ respectively represent the true frequency and the drift of the oscillator in the transmitter. While the receiver will set the frequency of its oscillator as f_n to down-convert y_{rf} , which is simplified as

$$\begin{aligned} y_S(m, n) &= y_{\text{rf}}(m, n) e^{-j(2\pi(f_r - f_n)t + w_r(t))} \\ &= x_S(m, n) e^{j(2\pi(f_t - f_r)t + w_t(t) - w_r(t))}, \\ &= x_S(m, n) e^{j(2\pi f_t t + w_t(t))} e^{-j(2\pi f_r t + w_r(t))}, \end{aligned} \quad (4)$$

where f_r and $w_r(t)$ respectively denote the true frequency and the drift of the oscillator in the receiver. It is worth noting that $\Delta\hat{\xi} = (f_t - f_r)$ is the CFO between the transmitter and the receiver. Generally speaking, for the same receiver, $\Delta\hat{\xi}$ is relatively consistent, while $\Delta\hat{\xi}$ between different receivers has large differences. This explains why there are obvious differences for different receivers shown in Fig. 1. Notably, $y_S(m, n)$ not only includes the CFO but also contains the oscillator drifts. To the best of our knowledge, most works consider $\Delta\hat{\xi}$ to be a confounding factor that needs to be compensated, without considering the characteristics of oscillator drifts. However, this paper finds that the oscillator drifts can become a significant RFF feature.

In practice, $\Delta\hat{\xi}$ is calculated from the STS by [5]

$$\Delta\hat{\xi} = \frac{1}{16} \angle \left(\sum_{m=6}^{M_S-1} \sum_{n=1}^{N_S} y_S(m, n) y_S^\dagger(m+1, n) \right), \quad (5)$$

where $(\cdot)^\dagger$ is complex conjugate. As a matter of fact, $\Delta\hat{\xi}$ is essentially the phase difference measured from the frequency responses of two identical and consecutive sequences. Specifically, for the i -th and k -th STS, they can be expressed by

$$x_S(k, n) = x_S(i, n) e^{j2\pi(k-i)N_S\hat{\xi}_i^k}. \quad (6)$$

If these two STSs are numerically identical, there is no CFO between them, i.e., $\hat{\xi}_i^k = 0$. Generally, the 10 STSs are exactly the same at the transmitter; but due to oscillator-induced drifts, there are certain phase differences in the received STSs. Hence, the cyc-similarity coefficient S_k^i can be calculated from the i -th and the k -th STSs as follows,

$$S_k^i = \frac{\sum_{n=1}^{16} y_S(i, n) y_S(k, n)^\dagger}{\sum_{n=1}^{16} |y_S(i, n)| |y_S(k, n)|}, \quad k > i. \quad (7)$$

Here, we do not perform angle calculation to obtain the specific $\hat{\xi}_i^k$, but directly use S_k^i that can retain the characteristics of the amplitude. Besides, the value of S_k^i ranges from 0 to 1; thus, no additional normalization is required.

Then, we analyze the impact of the drift $e^{-j(2\pi f_r t + w_r(t))}$ caused by different receiver oscillators on the designed feature. For the convenience of analysis, we denote the overall time-varying drift as $e^{-j f_r^t}$, and S_k^i can be represented by

$$S_k^i = \frac{\sum_{n=1}^{16} [y_S(i, n) e^{-j f_r^i}] [y_S(k, n) e^{-j f_r^k}]^\dagger}{\sum_{n=1}^{16} |y_S(i, n) e^{-j f_r^i}| |y_S(k, n) e^{-j f_r^k}|}, \quad (8)$$

where f_r^i and f_r^k are the drifts on the i -th and the k -th STSs, respectively. According to the experimental settings, the maximum time interval between the 1st STS and the 10th STS is $7.2 \mu\text{s}$. In such a short period of time, we consider that the receiver oscillator-induced drift will not change dramatically; hence, we can assume that $f_r^i \approx f_r^k$. Based on the property of the conjugate operation, (8) is simplified to (7), which means that the proposed S_k^i is robust to the drift of the receiver oscillator to a certain extent.

Considering the repeatability of STS and LTS, the extraction of the proposed cyc-similarity is depicted in Fig. 2. Note that

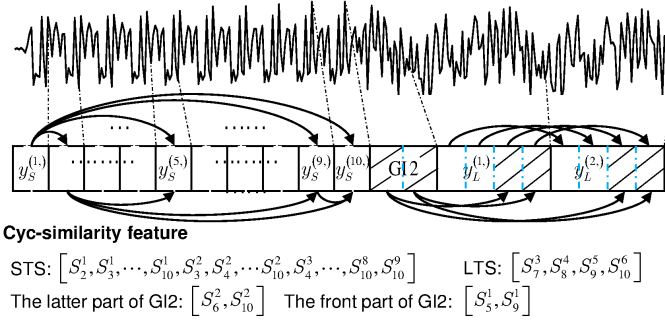


Fig. 2: The extraction of cyc-similarity feature.

GI2 is the cyclic prefix part of LTS. For the STSs, GI2, and LTSs in each preamble, there are 45, 4, and 4 cyc-similarity coefficients (53 in total) of complex-valued numbers, respectively. Indeed, these S_k^i constitute the oscillator drifts in a short period, which corresponds the variation of $w(t)$ in (1). Notably, the cyc-similarity exploits the repetitive nature of the signal, which can be easily found in preambles; thus, it can be extend to other devices as well. The 53 complex-valued S_k^i are split them into real and imaginary parts as $[\text{real}(S_k^i), \text{imag}(S_k^i)]$ to constitute a 1×106 vector; then, a simple KNN classifier is deployed to build an RFFI system.

IV. EXPERIMENTAL RESULTS

Note that the signals in WiSig are captured from 12 receivers on 4 days. For temporal and receiver generalization performance testing, our experiments only utilize signals collected from 1 receiver on 1 day to construct RFFI systems, which are then tested on the remaining signals. This can be considered the worst case. Here, as an example, we construct the RFFI system using the signal collected from the 1st receiver on the 1st day. For 6 WiFi devices, we utilize 900 signals per device (900×6 in total) to construct an RFFI system, and the remaining (100×6) signals are used for the same-day performance testing. For other receivers on different days, we utilize all ($1,000 \times 6$) signals for testing.

A. Comparison Method

As hardware defects are interrelated, it is difficult to represent them by simple handcrafted features [6]. Hence, most existing works adopt neural networks with feature mining capabilities to extract RFF features from signals automatically. For instance, authors providing WigSig utilize convolutional neural networks (CNN) to extract RFF features from the preamble [10]. A similar method can also be found in [6], [9]. For a more comprehensive comparison, we build three typical CNN-based RFFI systems that use the waveform, spectrum and spectrogram as inputs and select the well-verified ResNet18 for RFF extraction. Due to the maturity of this model, this paper will not elaborate on its parameters.

The training uses the stochastic gradient descent with momentum optimizer with a learning rate of 0.02 that drops every 20 epochs with a drop factor of 0.5. The L2 regularization factor is 10^{-4} and the mini-batch size is set to 64. The maximum number of epoch is 60, and the weights corresponding to the epoch with the minimum validation loss are reserved.

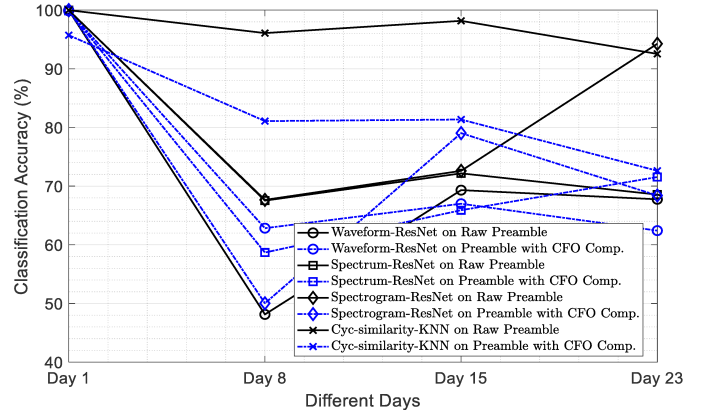


Fig. 3: Classification accuracies of different RFFI systems.

TABLE I: KNN with Different Number of Neighbors.

Neighbors	1	7	19	37	61	91
Accuracy	95.6%	95.6%	95.4%	95.2%	95.2%	95.0%

B. Temporal Generalization Performance

For different RFFI systems, the classification accuracies of signals captured on different days are shown in Fig. 3. Intuitively, for preambles without the CFO compensation, the proposed cyc-similarity-KNN achieves over 90% classification accuracy on different days. Interestingly, the classification accuracy decreases after compensating the CFO because the drift of CFO is destroyed.

As for RFFI systems consisting of ResNet18 and different signal representations, it is found that the systems cannot accurately identify signals on different days regardless of whether the CFO is compensated or not. It indicates that compensating the CFO does not necessarily guarantee an improvement in the identification performance. In fact, this result is mainly due to the following two reasons. On one hand, the impairment itself is imperceptible and time-varying; on the other hand, the wireless environment also makes the received signal time-varying. When only signals of a certain day are available, it is difficult for the CNN to extract RFF features that truly belong to devices. Instead, it will focus too much on the signals of this day. Thus, the system achieves accurately classify the signals captured on the same day, but produces disastrous results on other days.

For KNN classifiers, a significant parameter is the number of neighbors. Table I depicts the performance of systems with different number of neighbors. It is worth noting that all systems can achieve good performance, which means that the system is relatively robust to the choice of the number of neighbors. This also suggests that the designed cyc-similarity feature can well characterize hardware impairment features of devices, and thus, KNN can easily obtain the decision boundary.

C. Receiver Generalization Performance

For an RFFI system built using signals captured by a certain receiver, it is significant to investigate whether this system can

TABLE II: Classification Accuracy on Different Receivers.

RFFI systems	Classification accuracy of RFFI systems on different receivers												Average accuracy
	Rece.1	Rece.2	Rece.3	Rece.4	Rece.5	Rece.6	Rece.7	Rece.8	Rece.9	Rece.10	Rece.11	Rece.12	
Waveform-Res wo. CFO	71.2%	20.6%	37.8%	31.0%	26.6%	13.6%	19.7%	28.9%	44.5%	48.2%	36.0%	33.9%	34.4%
Waveform-Res w. CFO	75.8%	16.4%	45.2%	29.4%	51.8%	19.2%	17.7%	24.1%	45.9%	40.0%	41.5%	23.6%	35.9%
Spectrum.-Res wo. CFO	77.0%	29.5%	22.7%	15.1%	45.2%	30.4%	25.6%	37.2%	31.3%	35.9%	15.5%	30.4%	33.0%
Spectrum.-Res w. CFO	76.4%	18.5%	18.8%	16.0%	33.5%	27.2%	27.5%	33.9%	17.3%	31.8%	13.9%	34.1%	29.1%
Spectrogram.-Res wo. CFO	83.6%	20.0%	24.9%	0.8%	44.2%	14.4%	27.6%	18.0%	50.0%	33.3%	15.7%	20.1%	29.4%
Spectrogram.-Res w. CFO	76.4%	33.8%	29.3%	0.4%	47.8%	15.3%	18.1%	7.1%	55.4%	15.6%	16.3%	14.6%	27.5%
Cyc.-KNN wo. CFO	96.6%	80.6%	63.6%	38.1%	83.4%	74.0%	72.3%	82.2%	55.7%	57.2%	85.7%	62.9%	71.0%
Cyc.-KNN w. CFO	80.0%	58.0%	53.1%	29.4%	69.3%	29.8%	39.4%	42.7%	60.2%	47.9%	33.5%	48.8%	49.3%

be applied on other receivers, which can be viewed as receiver generalization performance. However, this case has not been fully considered as most literature utilizes only one receiver. Next, we deploy these RFFI systems on other receivers, *i.e.*, identify the signals captured by different receivers. The results are shown in Table II.

It is noted that the performance of the RFFI systems degrades when test signals come from different receivers. Indeed, the reason for this result is that the impairments of different receivers are different, and these impairments will inevitably be superimposed on the received signal. Specific to this paper, the oscillator of the receiver will have a certain impact on the received signal. For ResNet18, although it has strong feature extraction capabilities, it is difficult to learn effective RFF features directly from the signals of a certain receiver on a certain day. As a result, the ResNet18-based RFFI systems produce disastrous results on other receivers. Based on the analysis in Section-III, the proposed cyc-similarity feature can alleviate the impact of the receiver oscillator. Hence, the corresponding RFFI system archives a good performance on different receivers.

The above experiments verify the effectiveness of the proposed RFFI system. In terms of temporal generalization performance, the average classification accuracy of cyc-similarity-KNN reaches 96.6%. Compared with Spectrogram-ResNet18, the performance is improved by 16%. As for the receiver generalization performance, the average classification accuracy of cyc-similarity-KNN exceeds 70%, which is an increase of about 40%. Notably, we also conducted a number of other experiments, that is, randomly selecting the signals collected by a certain receiver on a certain day to construct RFFI systems. The same conclusions can be obtained.

Overall, this paper emphasizes that blindly relying on CNN may not be able to extract effective RFF features from signals, especially when signals are collected from only one receiver for one day. Besides, cautions should be exercised when performing signal processing. Although it eliminates some variations, impairment features may also be destroyed, such as the drift of oscillators. From the information level, we consider that the received raw signal already contains all impairment

features of devices. If effective RFF features can be extracted from the raw signal, the performance of RFFI systems can be further improved.

V. CONCLUSION

This paper proposed the cyc-similarity feature based on the repeatability of the preamble, which manifests the characteristics of the oscillator drifts. As the feature can characterize the device well, superior performance can be obtained using a simple KNN classifier. The designed RFFI system has good temporal and receiver generalization performance.

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